RESTRICTING A MORPHISM TO A CLOSED SUBSPACE GIVES A FIBRE SQUARE

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 697-704, following the statement at 672-695.

Editorial note. The sentence 'The rest of the lemma follows from the characterization in lemma-characterize-closed-subspace' covers f', its uniqueness and the fibre-square property; all are applications of that single published criterion (card L26388), flagged here.

Allowed standard prerequisites (editorial).

- Pullback of locally generated ideal sheaves. Modules, Lemma modules-lemma-pullback-locally-generated (cross-chapter, tag index verified, cited at ~699). J = Im(f^*I -> O_X) is locally generated by sections.
- Factoring through a closed subspace. lemma-characterize-closed-subspace (card L26388, cited at ~700-703). Supplies f', uniqueness and the universal property of the square.

Source reference. Original lemma 01HP: sources/stacks-schemes/pinned-original-source.tex, source lines 632–632. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01B3: focused-seeds/geometry/S9-schemes-restrict-map-to-closed/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $f : X \to Y$ be a morphism of locally ringed spaces. Let $\mathcal{I} \subset \mathcal{O}_Y$ be a sheaf of ideals which is locally generated by sections. Let $i : Z \to Y$ be the closed subspace associated to the sheaf of ideals $\mathcal{I}$. Let $\mathcal{J}$ be the image of the map $f^*\mathcal{I} \to f^*\mathcal{O}_Y = \mathcal{O}_X$. Then this ideal is locally generated by sections. Moreover, let $i' : Z' \to X$ be the associated closed subspace of X. There exists a unique morphism of locally ringed spaces $f' : Z' \to Z$ such that the following diagram is a commutative square of locally ringed spaces*

$$\begin{array}{ccc} Z' & \xrightarrow{i'} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ Z & \xrightarrow{i} & Y \end{array}$$

Date: stacks-project:schemes.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Moreover, this diagram is a fibre square in the category of locally ringed spaces.

Proof. The ideal $\mathcal{J}$ is locally generated by sections by Modules, Lemma 01B3. The rest of the lemma follows from the characterization, in Lemma 01HP above, of what it means for a morphism to factor through a closed subspace. □